Enhancing Software Quality in Research: Applying EVERSE insights to the B1MGplus project

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The role of software in scientific research has expanded significantly in the last years. As a result, measurement and enhancement of quality in research software has been a focal point amongst the Research Community. This poster aims to highlight dimensions, indicators and good practices in research software quality collected in the EVERSE project that could assist the B1MGplus (Beyond One Million Genomes Plus) project in building a sustainable software stack that can be deployed nationally, enhancing high-quality genomic research across Europe.

eosc | EVERSE

- **EVERSE** aims to create a community-driven framework that empowers researchers and developers to create, maintain, and share high-quality research software.
- Involves building a European network of Research Software Quality, laying the foundations of a future Virtual Institute for Research Software Excellence.
- **The Research Software Quality Kit (RSQkit)** integrates tools and services for software quality assessment, and engage with the research community through pilots, training, and recognition initiatives.

More evidence on **software quality indicators** in this recent paper. Some individuals who took part were funded by the EOSC EVERSE and ELIXIR STEERS projects.

ELIXIR STEERS

Promote good software management practices and support life scientists across Europe in managing software needs effectively.

Assessment of software components development, both **internal** (developers, using ELIXIR Software Management Plan) and **external** (users) to see the degree of finalization, integrity and usability of components.

IMPaCT Data

PH4GE

Promote best practices for public health bioinformatic pipelines enhancing the quality, reliability, and sustainability.

QUALITY DIMENSIONS: criteria relevant for assessing software quality

Performance Efficiency
Efficient performance under specific parameters and conditions

Flexibility
Adaptation to changes in requirements, context of use or system environment

Compatibility
Exchange information and/or share common environments/resources

Maintainability
Effectiveness/efficiency to modify a system/product to improve, correct, adapt it to changes in environment/requirements

FAIRness
Adherence of software to FAIR principles

Sustainability
Capacity to endure over time for new needs and platforms

Safety
Ability to prevent harm

Functional Suitability
Provide functions that meet stated/implied needs under specific conditions

Reliability
Function performance consistent under specific conditions and timeframe

Security
Ability to prevent attacks and ensure secure, authorized access to data by humans or machines

Interaction Capability
User interactions to exchange information and complete tasks across contexts

QUALITY INDICATORS: software aspect that can be measured

- **CodeMeta completeness (F):** % of properties filled with metadata (not the quality of metadata).
- **Existence of dependency management solution (Su):** Assesses external libraries/dependencies for compatibility and security.
- **Critical vulnerabilities (Se):** Checks if reported vulnerabilities have been fixed.
- **Leaked credentials (Se):** Detects hard-coded secrets (e.g. passwords) exposed in public git repositories.

(Please note this is a work in progress. At the time of creating this poster more indicators are being evaluated)

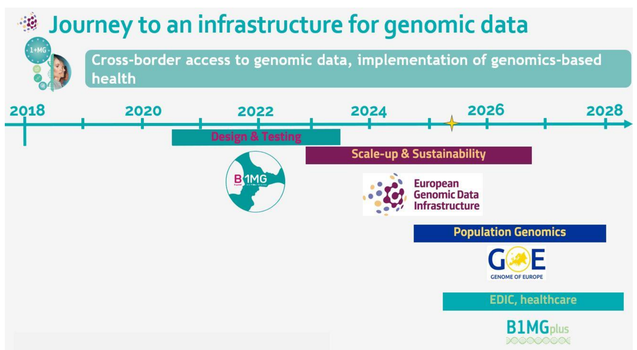
RECOMMENDATIONS

Tools to assess/improve software quality
RSQkit recommends different tools depending on **role** and **tasks** within B1MGplus.

Consider quality dimensions and indicators
e.g. FAIR principles for software.

Enhance software quality training & recognition
To promote reliability, sustainability of open-source software for genomic and health research.

- **B1MGplus** promotes secure, cross-border access to genomic and health data across Europe within the 1+ Million Genomes (1+MG) Initiative, in line with other projects such as GDI and GoE.
- Recognizes the need for indicators to assess research software quality and metadata, given the software complexity and collaborative development challenges.
- Interoperability among software components and standards for enabling federated analysis will also be considered.



B1MG plus

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